ALGEBRA 1 · EXPONENTIAL FUNCTIONS

Linear vs. Exponential Race

Explore & Reflect

Name _____ Date _____ Period _____

Open **Linear vs. Exponential Race** on Mr. K's Classroom: Algebra 1 → Exponential Functions → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. Starting at 2, which grows faster in the long run: adding 10 each step or doubling each step?

2. What do you think happens to $y = 2 \cdot b^x$ when b is between 0 and 1?

2 Explore use the tool and record what you see

Race: $y = 4x + 2$ and $y = 2 \cdot 2^x$

Set a = 2, b = 2, m = 4 and c = 2.

x	Linear y	Change	Exponential y	Change
0	2	—	2	—
1				
2				
3				
4				
5				

Change a and b in $y = a \cdot b^x$

a	b	y-intercept	Growth or decay?	Percent change per step
3	2			
5	0.5			
4	1.25			
2	1			

3 Write the rule fill in each blank from what you observed

1. A linear function changes by the same _____ each step. An exponential function changes by the same _____ each step.

2. In $y = a \cdot b^x$, a is the _____.

3. If $b > 1$ the function shows _____. If $0 < b < 1$ it shows _____.

4. A decay graph gets closer and closer to _____ but never reaches it.

5. An exponential growth function eventually _____ any linear function.

4 Check no tool: show your work

1. Linear or exponential: 5, 15, 45, 135?

2. Linear or exponential: 7, 11, 15, 19?

3. Describe $y = 100(0.8)^x$.

4. A town of 500 grows 10% per year. Write a model.

5 Reflect

How can you tell from a table whether a function is linear or exponential?